

JOSEPH MOSKOVITZ

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PERSONAL STATEMENT

Engineer with interests at the crossroads of space and nuclear technology, interested in the development & commercialization of nuclear power in the space domain in the long-term. Research experience in 3 different labs in 3 different countries, universities, and languages. Combined analytical engineering & finance knowledge & experience. Proficient in conducting literature reviews, designing and carrying out experiments, and authoring manuscripts. Ability to quickly learn and apply new software to relevant problems. Experience with project management, keeping team members motivated, assigning tasks, and ensuring completion of deliverables from myself and others. Excellent communication with supervisors and peers.

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY

Atlanta, GA

Master's in Science in Aerospace Engineering

May 2027

Cumulative GPA: 4.0/4.0

Specialized Coursework: Finite Element Analysis, Scientific Machine Learning

Bachelor's in Science in Aerospace Engineering

May 2026

Minor in Nuclear & Radiological Engineering

Cumulative GPA: 4.0/4.0; Dean's List 2022-2025

Specialized Coursework: Reactor Physics, Plasma Physics & Fusion Engineering, Radiation Protection Engineering, Space Policy

RANSOM EVERGLADES SCHOOL

Miami, FL

High School Diploma

Aug 2018 - May 2022

Cumulative GPA: 4.0/4.0; Cum Laude

RESEARCH EXPERIENCE

HIGH-POWER ELECTRIC PROPULSION LAB - GEORGIA TECH

Atlanta, GA

Undergraduate Research Assistant

Jan 2024 - current

- Machine Learning (ML) to Predict and Optimize Low-Power HET Performance Operating on Multiple Propellants
 - In collaboration with the Technion and previous work I conducted there, we are building an ensemble ML model called PCARE (Physics-Constrained Adaptive Residual Ensemble) with a multinational team (US, Israel, India) to predict a series of output parameters from raw data inputs as well as nonlinear features.
 - Performing multivariate optimization to optimize thrust to power (T/P) for different propellants (Xe, Kr, Ar, CO₂, N₂)
 - Writing a manuscript discussing ML model development and model inferencing for thruster performance optimization, targeting publication in the Journal of Electric Propulsion.
- Retarding Potential Analyzer (RPA) Redesign
 - Redesigned the lab's proprietary RPA housing, grids, and wiring connections to more accurately analyze different plasmas and to prevent shorting of different potential grids on older models.
 - Rebuilt CAD of individual parts and full assembly in SolidWorks, generated part drawings, GD&T with heavy DFM considerations due to the small and fragile nature of parts.
 - Created a BOM for the RPA, talked with various manufacturers and received quotes for different parts, machined parts in-house and ordered parts externally via lab management & procurement platform, and finally assembled 6+ fully functional units with interchangeable & spare parts.

EQUIPO DE PROPULSIÓN ESPACIAL Y PLASMAS - UC3M

Madrid, ES

Undergraduate Research Assistant

Sep 2024 - Dec 2024

- Developed a centralized machine/computer controls interface with LabVIEW for all PSUs involved in operating a HET.

ELECTRIC PROPULSION LABORATORY - TECHNION

Haifa, IL

Undergraduate Researcher

Jun 2024 - Aug 2024

- Characterized the performance and efficiency of the Simplified CAMILA low-power Hall-Effect Thruster (HET) across a diverse set of monatomic and molecular propellants (Xe, Kr, Ar, CO₂, N₂). Evaluated the viability of alternative propellants to mitigate industry reliance on Xe. Findings published at IEPC 2025 and currently under peer review at the Journal of Electric Propulsion.

WORK EXPERIENCE

REALTA FUSION

Madison, WI

Fusion Intern

May 2026 - Aug 2026

- Analyzing neutral beam injection (NBI) diagnostic data, including Doppler-shift and optical emission spectroscopy, to characterize beam performance on Realta's magnetic mirror fusion platform; building Python-based analysis pipelines.

ALLEN & COMPANY

New York City, NY

Investment Banking Summer Analyst

Jun 2025 - Jul 2025

- Worked on live deals with full-time analysts up to VPs in the Biopharma, SaaS, and sports industrial verticals. Leveraged valuation techniques (DCF, comps, precedents), M&A analysis (accretion/dilution), and heavy equities research.

GEORGIA TECH COLLEGE OF ENGINEERING

Atlanta, GA

Deformable Bodies Teaching Assistant

Jan 2024 - May 2024

- Created homework problems, held office hours to help students, proctored exams, and graded homework and exams for GT COE 3001.

PUBLICATIONS

J. Moskovitz, W. Brabston, D. Lev, M. L. R. Walker, M. Rubanovych, and V. Balabanov. "Performance Comparison and Analysis of Low Power Hall Thruster Operation on Inert and Molecular Propellants." *39th International Electric Propulsion Conference*, September 14-19 2025, London, UK. (IEPC-559)

PROJECTS

SCRAP-SAT

Atlanta, GA

Magnetics Engineer & Finance Lead

Jan 2026-current

- Developing an integrated satellite capable of detumbling and deorbiting tumbling, uncooperative space debris in Low-Earth Orbit (LEO) via magnetic eddy braking for detumble and attachment of a GOLD passive drag package for deorbit.
- Developed a Python (magpylib) design tool to optimize high-temperature superconducting (YBCO) pancake coils via multi-variable parameter sweeps to maximize magnetic field strength on the target debris.
- Engineered a MATLAB Monte Carlo simulation of magnetic eddy-current detumbling to quantify rotational energy dissipation and mission duration across a broad trade space of debris properties.
- Developing multi-year pro forma financial models and an investor pitch for a potential startup spinout.

1-D REACTION WHEEL

Atlanta, GA

Project Lead

Jan 2026-current

- Designing and constructing a reaction/momentum wheel testbed to demonstrate spacecraft attitude control; 3D printing the structure and integrating the motor, Arduino, IMU, and power electronics on the satellite disk.
- Modeling open- and closed-loop system behavior in Simulink and designing a PID attitude controller.

COMMUNITY LEADERSHIP

CHABAD ON CAMPUS

Atlanta, GA

Executive Board Member

Jan 2024 - current

- Plan holidays, parties, social events, and more for the Jewish community at Georgia Tech
- Increased the number of events for each Jewish holiday and helped increase weekly Shabbat attendance by over 50% in 3 years

ZETA CHAPTER OF ALPHA EPSILON PI

Atlanta, GA

Exchequer

Apr 2023 – Apr 2025

- Managed the finances of the fraternity, including collecting dues, filing taxes, paying bills, and setting the budget
- Incorporated the Chapter as a 501(c)(7) nonprofit organization, acquired a line of credit in the Chapter's name and implemented new expense-tracking software, authored and passed an organization Expense Policy

ADDITIONAL

Technical Skills: SolidWorks, Python, MATLAB, Simulink, LabVIEW, Finite Element Method Magnetics (FEMM), Data Analysis, Machine Learning, Excel, Woodworking, 3D Printing, Circuit design & construction

Languages: Bilingual in English & Spanish, Conversational in Hebrew